34. Beamforming Signal Rate Conversion

Aim:

To simulate the sampling rate conversion of a beamformed signal from 61.44 MHz to 30.72 MHz and verify the preservation of phase alignment and beamforming integrity.

Objectives

1. Convert the sampling rate from 61.44 MHz to 30.72 MHz using decimation.
2. Analyze the phase alignment before and after conversion.
3. Measure amplitude distortion.
4. Compare the input and output spectra.
5. Verify that beamforming performance is preserved.

Output:

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AMPLITUDE ANALYSIS
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Amplitude before = 1.000000
Amplitude after  = 1.000925
Amplitude distortion = 0.092517 %

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BEAMFORMING INTEGRITY
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Input antenna power = 1.001851
Beamformed power = 1.001851
Beamforming gain = 0.000 dB

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PERFORMANCE SUMMARY
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Input Sampling Rate  : 61.44 MHz
Output Sampling Rate : 30.72 MHz
Rate Conversion      : Decimation by 2
Maximum Phase Error  : 0.000000 degrees
Amplitude Distortion : 0.092517 %
Beamforming Gain     : 0.000 dB
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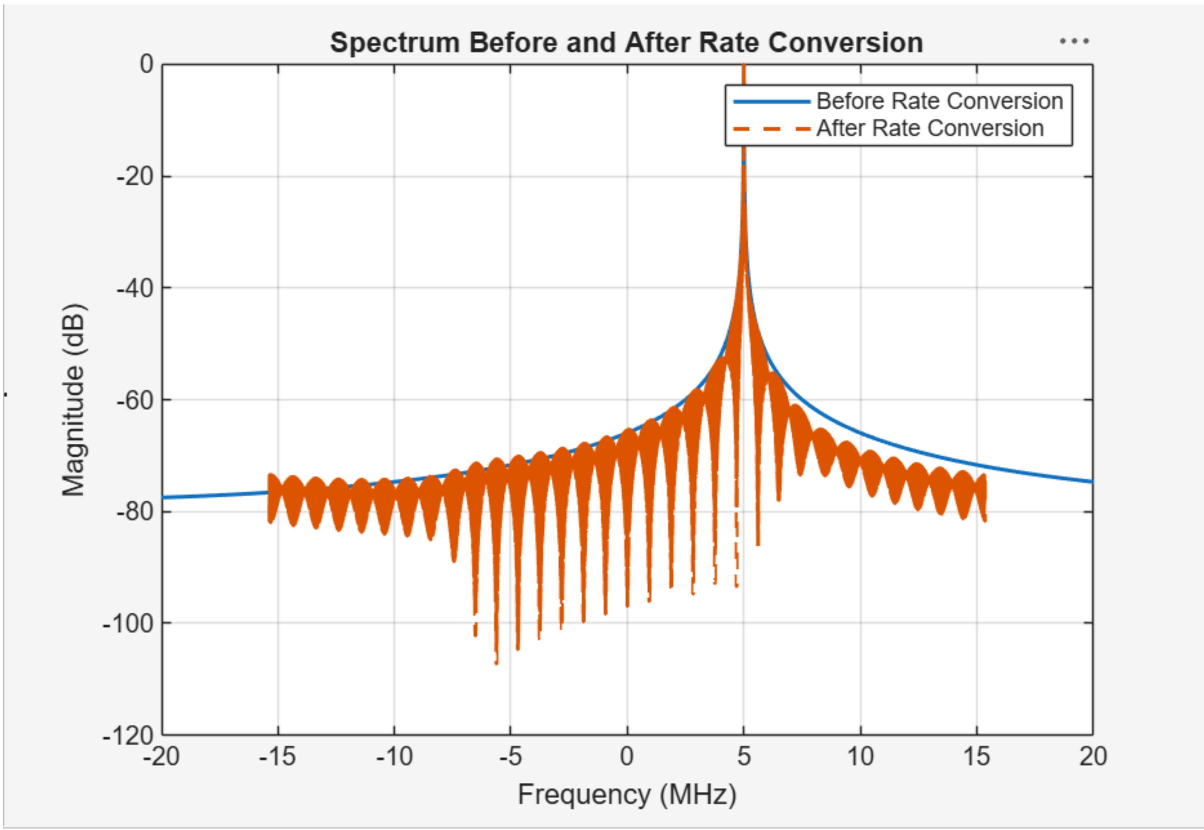
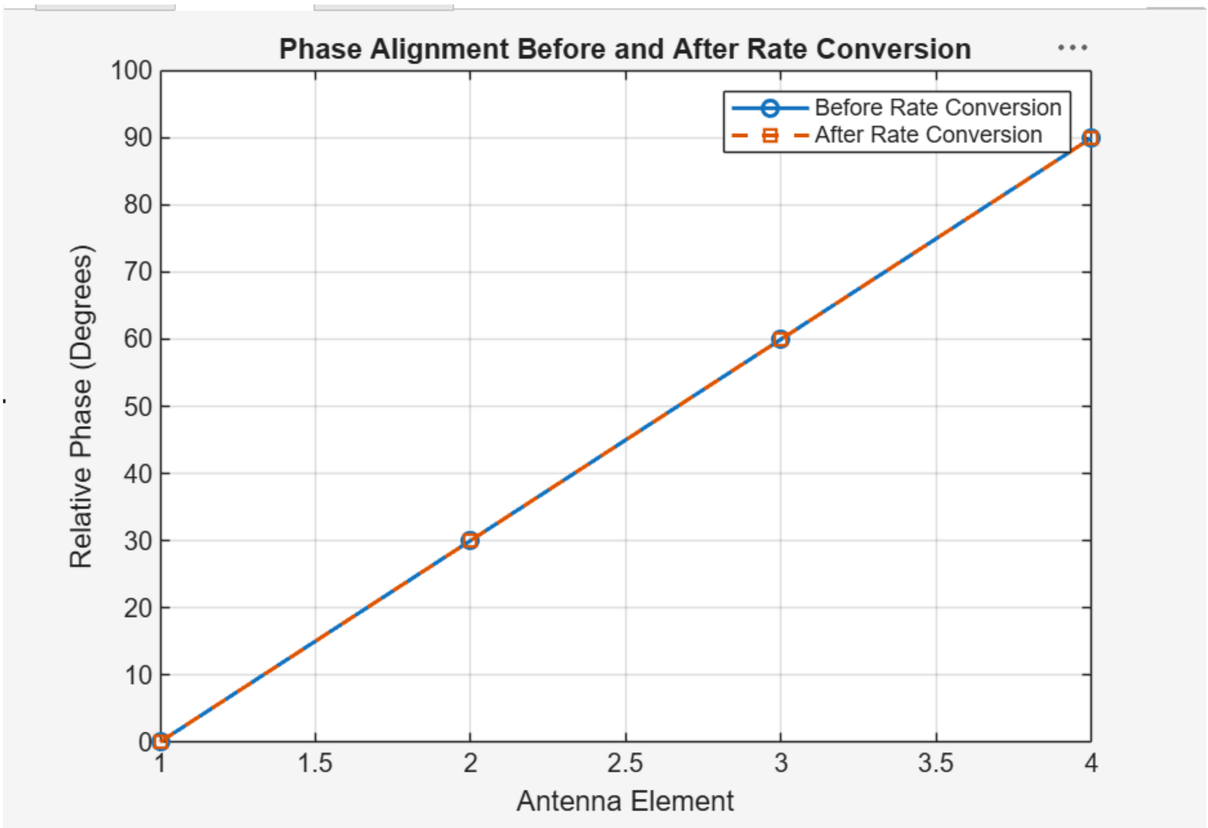
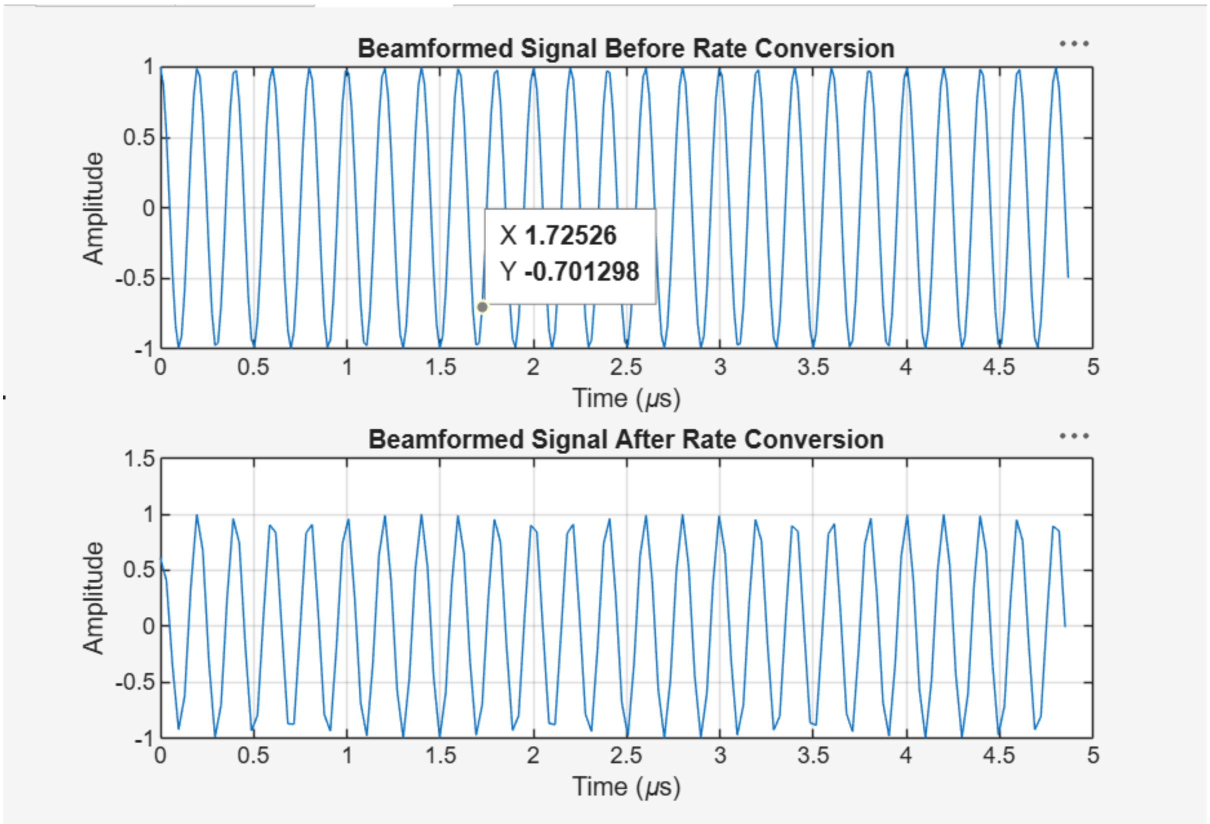
```


Spectrum Before and After Rate Conversion
Before Rate Conversion
After Rate Conversion
Magnitude (dB)
0
-20
-40
-60
-80
-100
-120
-20
-15
-10
-5
0
5
10
15
20
Frequency (MHz)


Phase Alignment Before and After Rate Conversion
Before Rate Conversion
After Rate Conversion
Relative Phase (Degrees)
100
90
80
70
60
50
40
30
20
10
0
1
1.5
2
2.5
3
3.5
4
Antenna Element



Result:

The beamformed signal was successfully converted from 61.44 MHz to 30.72 MHz using decimation by 2. The simulation showed minimal amplitude distortion, preserved phase alignment, and maintained the desired spectral and beamforming characteristics after rate conversion.